Configuration Checklist - What You Need to Change

1. Find Your Raspberry Pi IP Address

First, on your Raspberry Pi, run:

```bash
hostname -I
```

Example output: `192.168.1.100`

Write it down: _____

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2. Update the Web Interface (index.html)

File: `static/index.html`

Line 20 - Change this:

```javascript
const API_BASE_URL = 'http://192.168.1.100:5000';  // UPDATE THIS!
```

To:

```javascript
const API_BASE_URL = 'http://YOUR_RASPBERRY_PI_IP:5000';
```

Example:

```javascript
const API_BASE_URL = 'http://192.168.1.150:5000';
```

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3. Check Your ROS2 Distribution

On Raspberry Pi, check which ROS2 version you have:

```bash
echo $ROS_DISTRO
```

Common versions:

- humble (recommended)
- foxy
- galactic

Write it down: _____

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4. Update server.py ROS2 Commands (if needed)

File: server.py

If your ROS2 topics or action names are different, update these sections:

Velocity Topic (around line 30-40)

Current:

```python
'/cmd_vel'
```

Check your actual topic name:

```bash
ros2 topic list | grep vel
```

Dock Action (around line 100)

Current:

```python
'/dock'
'irobot_create_msgs/action/Dock'
```

Verify:

```bash
ros2 action list
ros2 action info /dock
```

Undock Action (around line 115)

Current:

```python
'/undock'
'irobot_create_msgs/action/Undock'
```

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5. Source Your ROS2 Workspace

Before running server.py, you MUST run these commands:

```bash
# Source ROS2 (change 'humble' to your version)
source /opt/ros/humble/setup.bash

# Source your workspace
source ~/create3_examples_ws/install/local_setup.sh

# Set safety override
ros2 param set /motion_control safety_override full
```

Your workspace path might be different! Check if you have:

- `~/create3_examples_ws/`
- `~/ros2_ws/`
- `~/catkin_ws/` (ROS1)
- Something else?

Write it down: _____

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6. Optional: Change Server Port

File: `server.py`

Last line - If port 5000 is taken, change:

```python
app.run(host='0.0.0.0', port=5000, debug=True, threaded=True)
```

To:

```python
app.run(host='0.0.0.0', port=8080, debug=True, threaded=True)
```

Then also update index.html:

```javascript
const API_BASE_URL = 'http://YOUR_IP:8080';
```

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7. Firewall Configuration (if needed)

If you can't connect from browser:

```bash
# Allow the port through firewall
sudo ufw allow 5000

# Or disable firewall temporarily for testing
sudo ufw disable
```

Summary - Files to Modify:

✅ REQUIRED Changes:

1. `static/index.html` - Line 20
   - [ ] Update `API_BASE_URL` with your Raspberry Pi IP

🔧 MIGHT NEED Changes:

2. **Terminal (before running)** - Source commands
   - [ ] Source correct ROS2 distribution
   - [ ] Source correct workspace path
   - [ ] Set safety override

3. `server.py` - Only if your setup is different
   - [ ] Check topic names match (`/cmd_vel`)
   - [ ] Check action names match (`/dock`, `/undock`)
   - [ ] Change port if 5000 is taken

Quick Test Commands

After making changes, test these on Raspberry Pi:

```bash
# 1. Check ROS2 is working
ros2 node list

# 2. Check topics exist
ros2 topic list

# 3. Check actions exist
ros2 action list

# 4. Test manual movement
ros2 topic pub --once /cmd_vel geometry_msgs/msg/Twist "{linear: {x: 0.2}}"

# 5. Test dock action
ros2 action send_goal /dock irobot_create_msgs/action/Dock "{}"
```

If these work, your configuration should be correct!

Still Not Working?

Check the setup instructions artifact for detailed troubleshooting steps.